Review

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Theorem 0.1 (Implicit function theorem): If the mapping $F : U \to \mathbf{R}^n$ defined in a neighbourhood U of the point $(x_0, y_0) \in \mathbf{R}^{m+n}$ is such that

1. $F \in C^p(U; \mathbf{R}^n), p \geq 1$;
2. $F(x_0, y_0) = 0$;
3. $F'_y(x_0, y_0)$ is an invertible matrix.

Here we set

$$F'_x(x,y) = \begin{pmatrix} \frac{\partial F^1}{\partial x^1} & \dots & \frac{\partial F^1}{\partial x^m} \\ \vdots & \ddots & \vdots \\ \frac{\partial F^n}{\partial x^1} & \dots & \frac{\partial F^n}{\partial x^m} \end{pmatrix}(x,y),$$

$$F'_y(x,y) = \begin{pmatrix} \frac{\partial F^1}{\partial y^1} & \dots & \frac{\partial F^1}{\partial y^n} \\ \vdots & \ddots & \vdots \\ \frac{\partial F^n}{\partial y^1} & \dots & \frac{\partial F^n}{\partial y^n} \end{pmatrix}(x,y).$$

Then there exists an $(m+n)$-dimensional interval $I = I_x^m \times I_y^n \subset U$, where

$$I_x^m = \{x \in \mathbf{R}^m : |x - x_0| < \alpha\}, \quad I_y^n = \{y \in \mathbf{R}^n : |y - y_0| < \beta\},$$

and a mapping $f :\in C^p\left(I_x^m; I_y^n\right)$ such that

$$F(x,y) = 0 \Leftrightarrow y = f(x),$$

for any point $(x,y) \in I_x^m \times I_y^n$ and

$$f'(x) = -\left[F'_y(x, f(x))\right]^{-1}\left[F'_x(x, f(x))\right].$$

Theorem 0.2 (Taylor formula): If the function $f : U(x) \to \mathbf{R}$ is defined and belongs to class $C^{(n)}(U(x); \mathbf{R})$ in a neighborhood $U(x) \subset \mathbf{R}^m$ of the point $x \in \mathbf{R}^m$, and the closed interval $[x, x + h]$ is completely contained in $U(x)$, then the following equality holds:

$$\begin{aligned} &f(x^1 + h^1, ..., x^m + h^m) - f(x^1, ..., x^m) \\ &= \sum_{k=1}^{n-1} \frac{1}{k!} (h^1 \partial_1 + \cdots + h^m \partial_m)^k f(x) + r_{n-1}(x; h), \end{aligned} \tag{1}$$

where

$$r_{n-1}(x; h) = \int_0^1 \frac{(1-t)^{n-1}}{(n-1)!} (h^1 \partial_1 + \cdots + h^m \partial_m)^n f(x + th)\, dt. \tag{2}$$

Equality Equation 1, together with Equation 2, is called *Taylor's formula with integral form of the remainder.*

Remark. We have the equality

$$\begin{aligned} &f(x^1 + h^1, ..., x^m + h^m) - f(x^1, ..., x^m) \\ &= \sum_{k=1}^{n} \frac{1}{k!} (h^1 \partial_1 + \cdots + h^m \partial_m)^k f(x) + o\Big(\parallel h \parallel^n \Big) \quad \text{as} \quad h \to 0, \end{aligned} \tag{3}$$

called *Taylor's formula with the remainder term in Peano form.*